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Anebilin

Non-invasive spectral blood biomarker scanner
for fast preliminary screening

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PROJECT PROTOTYPE

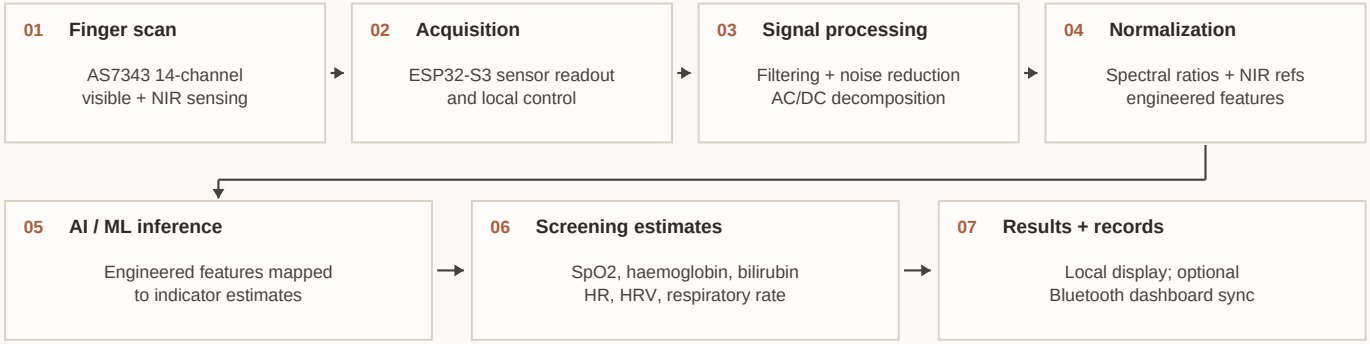
PROBLEM & OBJECTIVE

In resource-constrained communities, timely screening can be limited by reliance on blood collection, laboratory facilities and trained personnel. Anebilin is a portable finger-scan prototype intended to estimate multiple physiological indicators non-invasively for preliminary screening and monitoring.

METHODOLOGY

An AS7343 14-channel sensor captures visible and near-infrared (NIR) optical signals from a finger. An ESP32-S3 acquires the readings and performs filtering, noise reduction and AC/DC decomposition. Spectral ratios and NIR reference channels support normalization against optical variation; engineered features are processed by machine-learning models to estimate SpO2, haemoglobin, bilirubin-related indicators, heart rate, HRV and respiratory rate. Results appear locally, with optional Bluetooth synchronization to a dashboard for records and longitudinal monitoring.

PROPOSED SYSTEM BLOCK DIAGRAM



Optical signals are transformed into engineered features before model inference; outputs are screening estimates.

PROPOSED OUTCOME

Deliver an integrated portable prototype that combines multispectral sensing, on-device signal processing, machine-learning-based indicator estimation and connected records. The goal is to provide rapid preliminary insights from a single finger scan and support longitudinal monitoring through the dashboard.

PROJECT VIDEO

Anebilin by Team Penteract - watch the project demo

youtu.be/OL798aeKOY0